PRD · AI Learning Platform for Children (5–17)

Cohort 8 · Case Study 4 · Author: Aditya Gupta · 26 Aug 2026 · v2, 30 Aug 2026

Evidence legend — [V] verified against a primary source · [S] third-party/analyst estimate, attributed not asserted · [P] our own primary interviews · [A] our assumption, open to challenge. Every load-bearing claim carries a tag.

1. Problem Framing & Focus

1.1 Case Summary & Given Problem

The brief pointed at non-technical professionals struggling with AI. We interviewed working professionals across teaching, sales, operations and IT, plus school children and their parents [P]. Every adult conversation converged on the same thing: the struggle with AI is not a tools problem, it is a thinking problem that was set years before they entered the workforce. Schooling optimises for execution — memorise the method, reproduce it in the exam. Nobody is asked to frame a problem, weigh a trade-off, or judge whether an answer is any good. Those are exactly the skills AI now demands of its user.

Meanwhile the offloading is measurable. A 2025 study of 666 UK participants found frequent AI-tool use negatively correlated with critical-thinking scores ($r = -0.68$), mediated by cognitive offloading, with the youngest band (17–25) showing the highest dependence and the lowest critical thinking (Gerlich, Societies 15(1):6) [V — cross-sectional self-report, association not causation].

Key Insight: *Adults struggling with AI are the symptom. The cause is an education system that never built the thinking layer under the execution layer. Retraining a 35-year-old is remediation; reaching a child is prevention.*

1.2 Narrowing the Buyer: Indian parents already paying for preventive cognitive skilling

The pain in 5–17 is felt by the child but bought by the parent, and that buyer already exists. Indian parents demonstrably pay, years ahead of any payoff, for cognitive skills their child "might not need" — abacus, mental arithmetic, kids-coding. Posted market prices run ₹600–1,200/month for group abacus and ₹2,000–2,500/month for branded abacus or 1-on-1 coding [V, vendor price lists]. NEP 2020 recommends coding activities from the Middle Stage, Grades 6–8 [V, NEP 2020 ¶4.25/¶4.4], which has socially normalised "future-skills" spend. We are not creating a budget line; we are redirecting one.

The parent is also already anxious and already blind. Common Sense Media (2024) found 7 in 10 US teens have used a generative-AI tool and 53% use it for homework help, but only 37% of parents whose teen used one knew about it, and 49% had not discussed AI with their child at all [V]. Our own parent interviews name the fear precisely: "Earlier english skills were great, now dependency. She has less thoughts on her own." [P]

Key Insight: *The buyer is the parent of a 5–17 child who already pays for preventive cognitive skilling and already fears AI dependency — high pain, proven willingness to pay, no product serving it.*

1.3 Narrowing the Domain: AI-thinking foundations, not homework help

We deliberately do not compete on getting schoolwork done. Homework help is the exact mode that produces offloading, it is a crowded free category, and it is the behaviour parents told us they are trying to stop [P]. We narrow to the thinking layer: framing a problem, predicting, judging an output, and repairing a wrong mental model — taught through AI as the subject matter.

The policy tailwind confirms this is the contested skill. Beijing became the first province-level region in China to make AI education compulsory from the September 2025 semester, across 1,400+ schools with at least eight class hours per grade, part of a national goal to universalise AI education by 2030 — and its guidelines bar primary-school children from independently using open-ended generative AI [V]. The US signed Executive Order 14277, "Advancing Artificial Intelligence Education for American Youth", on 23 April 2025 [V]. OECD PISA 2022 Creative Thinking (published 2024) showed creative thinking is measurable and system-dependent: Singapore 41, Korea and Canada 38, OECD average 33 [V].

Key Insight: *Both AI superpowers reached the same conclusion — teach children to understand AI before letting AI do their thinking. That is the product.*

2. Discovery

2.1 Research Methodology

Primary: 6 completed interviews with children in 5th grade (ages 9–11) and children who had just completed 10th grade, plus their parents; a Google Forms survey; and professional interviews across teaching, sales, operations and IT [P]. Teacher interviews are still pending. Secondary: a 15-document internal research bundle (docs/grill-me) built on UNICEF, IES, APA, Brookings, W3C, Common Sense Media and OECD sources, plus market data verified against UDISE+, MoSPI and NEP primary documents.

2.2 Primary Research — three findings that set the problem

#	Finding	Verbatim evidence [P]
1	Dependency — not access, not jobs — is the number one parent fear.	"Surface level knowledge from AI. Mastery of any subject doesn't lie in the gist." · "The ability to calculate in mind has come down." — Parents
2	Children can already articulate their own offloading, better than adults can.	"Nobody uses a calculator or clock anymore. We are only changing who we are giving the input to." · "The latest phones have daily briefs. We just want the headlines. That inquisitiveness goes away." — Students
3	Children love building, but only inside games — and the first real build fails and ends there.	"Minecraft — dream, create… make your own rules." (5th grade) · "I told AI to give me the full code. It went horribly wrong. That was 2 years ago." (never tried again)
4	School is not filling the gap.	Interviewees described computer class as WordArt in MS Word, block coding "not the best", 150 marks theory vs 50 practicals. The one child genuinely building taught himself outside school.

2.3 Secondary Research

Evidence	Finding	Tag
Gerlich, Societies 15(1):6 (2025)	666 participants; AI use ↔ critical thinking r = −0.68, mediated by cognitive offloading; 17–25 worst affected. A Sept-2025 correction fixed a duplicated table; conclusions unaffected.	[V]
Sparrow et al., Science 333:776 (2011)	"Google effect": with expected future access, people recall where to find a fact (M=0.49) over the fact itself (M=0.23). Influential but a known replication target.	[V]
Common Sense Media (Sept 2024)	7/10 teens used a genAI tool; 53% for homework help; only 37% of parents knew; 49% had never discussed it. n=1,045 US parent–teen pairs.	[V]
IES Practice Guide / APA Practice	Retrieval practice, spacing, worked-example fade-out and action-specific feedback produce durable learning; vague praise does not.	[V]
Brookings, Realizing the Promise	Edtech fails when it replaces effective instruction, distracts, or adapts too weakly to learner need.	[V]
UNICEF, Policy Guidance on AI for Children	AI for minors must be bounded, adult-in-the-loop, and data-minimising; safeguarding is a design requirement, not a compliance annex.	[V]
OECD PISA 2022 Creative Thinking	Singapore 41, Korea/Canada 38, OECD avg 33, 64 systems. Creative thinking is trainable and curriculum-driven, not a fixed trait. US opted out.	[V]
Spherical Insights / market.us	Online coding-for-kids ≈ US$6.5B (2025) → US$32B (2035), ~17% CAGR; AI in K-12 ~37% CAGR off a sub-US$0.5B base. Vendor estimates, 14–18.6% CAGR spread.	[S]

Key Insight: *The threat is measurable, the adoption is already unsupervised, and the pedagogy that counters it — retrieval before explanation — is well evidenced. The market data is the weakest layer and is attributed, not asserted.*

2.4 Current Journey: how a child meets AI today

#	Step	Where it breaks
1	Child hits a task they cannot do — homework, a build idea, a question.	No framing step. The task is never decomposed.
2	Child opens a general chatbot and asks for the	Prompting itself is a barrier: "It takes 2–3 minutes to

	answer.	come up with the right prompt." [P]
3	AI returns a complete, fluent answer.	The thinking is done before the child attempts anything. Offloading happens here.
4	Child copies it, submits, moves on.	No retrieval, no explanation, no evidence of understanding. Completion masquerades as mastery.
5	Parent sees a finished task and a good mark.	63% of parents never see step 2 at all [V, Common Sense 2024]. No signal, no intervention point.
6	Habit hardens.	By 13–17 dependency is self-reported by the child and visible to the parent [P].

Key Insight: *The break is steps 2–4: the child outsources the thinking before ever attempting it, and no adult sees the moment it happens.*

2.5 Stakeholder Mapping

Stakeholder	**Role**	**Cares about**
Learner 5–7	Plays, taps, chooses; cannot yet read or type fluently	Fun, short steps, immediate response
Learner 8–12	Practises, retries, builds small things	Feeling capable; "I made this"
Learner 13–17	Explains, compares, builds projects	Autonomy, real output, not being talked down to
Parent / Guardian	Buys, consents, monitors	Proof the child is thinking, not copying; safety; time cost
Teacher (pending)	Assigns, observes classroom effect	Whether AI is helping or hollowing out their students
Safety reviewer	Reviews escalated risk events	Precision of alerts; nothing severe missed
Content author	Writes lessons without engineering help	Being able to ship a lesson as a validated content package

2.6 From interview to feature — direct traceability

Every capability below exists because a parent or child said something in an interview. Three of the six surfaced only because we let parents talk past the questions we had prepared. [P]

What we heard [P]	What it told us	What it became
"Teach him AI, but don't let AI think for him."	The parent wants AI literacy and fears AI dependency in the same sentence.	The retrieve-then-guide loop: the child attempts, the system explains only afterwards, then asks for recall or transfer. Explanation never precedes the attempt.
"I don't want my child chatting with a bot."	A general chat surface is itself the objection, independent of what the bot says.	No open chat anywhere in the product. Every server-side AI call passes through one gated library; input and output are both moderated.
"Only parents can register on the school app — but it is the grandparents who teach him."	The two-guardian assumption in school software excludes the adults doing the teaching.	A family holds many guardians with roles, not a fixed mother-and-father pair. Any guardian can follow a child's progress; the head guardian holds the consent decisions.
"She studies only when it feels like play."	Attention is the constraint before pedagogy is.	Subjects are worlds, topics are island maps, lessons are stages that unlock in order, with XP and badges tied to mastery progress rather than to time on app.
"Cartoons move too fast for her."	Ambient motion is a comfort and attention problem for some children, not a delight.	Every ambient animation is gated behind the operating system's reduced-motion setting. The product reads the device preference; it never asks the child.
Split: some families wanted a community of learners, others wanted privacy.	There is no single right answer, so the product must not pick one for the family.	Two explicit choices. A child signs in with an internal skooling.in username that has no inbox, or (ages 13–17) a real Gmail. A shared picture goes out named or anonymous, chosen by the guardian at approval time; anonymous is the default.

Key Insight: *The three findings we did not plan for — the guardian structure, the attention constraint and the motion sensitivity — each produced a feature. The research method that mattered was letting the interview run past its script.*

3. User Personas

Three age bands, taken unchanged from the product's confirmed decision log (docs/grill-me/12-decisions.md, 2026-08-21): one flow for 5–17 is too broad for reading load, motor skill and autonomy.

Band	Role in learning	Core pain	Primary need	Input & support bias
3.1 5–7 Early readers	Guided story-play. Consumes screens; creates on paper, not on screen.	Cannot read or type well enough for any chat-based AI; text interfaces exclude them entirely.	Guided story-play with short retrieval checks and no reading cliff.	Tap, drag, image choice, very low typing. Optional spoken instructions, never forced narration.
3.2 8–12 The forming years	Challenge-practice. Fluent user of technology but almost entirely as a consumer; building happens only inside games.	First independent AI exposure begins here, unsupervised, and the habit sets in copy-the-answer mode. Habits are not yet hardened — this is the intervention window.	Short daily loop that visibly builds thinking, guided builds that always end in something that works.	MCQ, short bounded text, guided builds, explicit retry loops. Medium scaffolds, clear progress cues.
3.3 13–17 The hardened years	Project-plus-reflection. AI already embedded in schoolwork.	Dependency is visible and self-reported; first solo build failed and killed motivation; board-exam pressure crowds out everything.	Real projects and reasoning tasks that respect autonomy; a reason to try building again.	Text, comparison, critique, simulations, projects. Higher autonomy, strategic hints, deeper reflection.
3.4 Parent / Guardian	Buyer, consent-giver, monitor. Not present during use.	Fears dependency, cannot see the offloading happen, and has no evidence of whether learning occurred.	Proof of thinking — not activity minutes — plus honest safety reporting and a short time commitment.	End-of-day summaries, mastery evidence, urgent alerts only when warranted.

Persona jobs-to-be-done

Persona	"..."
5–7	"Let me play the story and show you I got it, without making me read a wall of text."
8–12	"Help me make something real that works, and show me I am getting smarter."
13–17	"Give me a real problem worth my time, and do not do the thinking for me."
Parent	"Show me my child is thinking, not copying — and tell me fast if something is wrong."

4. Pain Points & Prioritisation

4.1 Problem Statements

#	Problem	Research finding	Tag
P1	Cognitive offloading	Frequent AI use correlates with lower critical thinking, mediated by offloading; the youngest cohort is worst affected.	[V]
P2	Unsupervised, unstructured first exposure	Majority teen adoption with only 37% parent awareness; the habit is set in copy-the-answer mode before anyone intervenes.	[V]
P3	Completion is mistaken for mastery	Finished content does not prove understanding; one correct answer overstates it (APA, IES).	[V]
P4	One flow cannot serve 5–17	Reading load, motor skill and autonomy differ so sharply that a shared flow fails all three bands.	[V]
P5	The first real build fails and ends there	"It went horribly wrong. That was 2 years ago." — never tried again. Games give the building pride; nothing bridges game-building to real building.	[P]
P6	School does not fill the gap	WordArt, block coding, 150 marks theory vs 50 practicals; the only genuine builder taught himself outside school.	[P]
P7	Parents have no signal	No visibility into whether the child thought or copied; dependency only becomes visible years later.	[P/V]
P8	Family software assumes two parents	A parent described a school app that registers only a mother and a father, while the grandparents do the daily teaching. The adults doing the work cannot see the record.	[P]
P9	Families disagree about visibility	Interviews split on community: some children and parents wanted to see and be seen by other learners, others wanted no exposure at all. A single default fails one group.	[P]

Key Insight: *Every problem shares one structural gap: children are being handed finished answers by a system with no way to require thinking first and no way to prove thinking happened. P8 and P9 are different in kind — they are about the family around the child, and both became product decisions rather than settings.*

4.2 Problem Prioritisation — MVP focus: Thinking Evidence + Bounded AI

Criterion	Thinking Evidence (visibility layer)	Bounded AI (risk-control layer)
Frequency	Every concept, every learner, every session.	Every AI touchpoint in the product.
Severity	Without it, the product cannot distinguish a child who understood from one who copied — and the parent cannot either.	An open chat tutor reproduces the exact offloading and safety exposure we exist to prevent.
Root cause	Mastery is inferred from completion instead of from performance plus explanation or transfer.	General-purpose AI answers first and is unbounded in topic, tone and risk.
MVP value	Makes learning provable to the learner, the parent and the routing engine.	Makes AI safe and pedagogically correct by construction, not by policy.
Serves	P1, P3, P7	P1, P2, P5

Why these two

Test	Argument
Universal	Both apply to all three bands and every lesson. Nothing else in the list does.
High impact	Together they attack offloading directly — the one harm that is both measured in the literature and named unprompted by our parents.
Compounding	Without mastery evidence, bounded AI is unprovable; without bounded AI, mastery evidence is contaminated by AI-authored answers.
Foundational	Once thinking is provable and AI is bounded, the rest — projects, gamification, teacher tools, more subjects — can be built on top safely.
Deliberately deferred	P4 is a solved design decision (three bands, already confirmed), P6 is context not a product surface, and P5 is downstream of the two we chose.

Key Insight: *Thinking Evidence is the visibility layer; Bounded AI is the risk-control layer. Together they are the minimum that makes an AI product for children defensible to a parent.*

5. Competitive Landscape & Gap

Directional positioning based on public product descriptions — verify each row before submission [A].

Category	Examples	What they solve	Key gap
General AI assistants	ChatGPT, Gemini, Claude	Any question, instantly, free	Answer-first by design. No pedagogy, no age banding, no mastery evidence, no parent visibility. This is the offloading engine.
AI homework helpers / tutors	Khanmigo, Photomath, Gauth	Subject-specific help on set schoolwork	Optimise for task completion, which is the behaviour parents want stopped. Mastery still inferred from completion.
Kids coding & STEM academies	Tynker, Code.org, coding-class chains	Structured curriculum, real builds, live instruction	Teach execution (syntax, blocks), not framing and judgement. Expensive, scheduled, instructor-bound; not a short daily loop.
Preventive cognitive skilling	Abacus & mental-arithmetic chains	Proven parent willingness to pay for a cognitive skill with a delayed payoff	Trains a skill AI already automates. Offline, no adaptivity, no AI literacy.
Gamified learning apps	Duolingo, Khan Academy Kids	Habit, streaks, delightful age-appropriate UX	Streak-first motivation; rewards track activity rather than mastery, which is the failure mode we are explicitly designing against.

5.2 The Gap

Bounded AI + Age-banded adaptive routing + Provable thinking evidence — in one product, for 5–17.

Today a parent has to assemble it themselves: a chatbot the child misuses → a coding class that teaches syntax → an abacus class that trains an obsolete skill → and no evidence at any point that the child is thinking better. Nobody owns the outcome.

Capability	General AI	Homework helpers	Coding academies	Abacus	Proposed
Retrieval before explanation	✗	✗	Partial	✓	✓
Mastery = correct + explain/transfer	✗	✗	Limited	✗	✓
Misconception-targeted repair	✗	Limited	✗	✗	✓
Age-banded interaction model (5-7/8-12/13-17)	✗	Limited	Limited	Limited	✓
Bounded AI, no open chat surface	✗	✗	n/a	n/a	✓
Layered safeguarding + parent alerts	Limited	Limited	n/a	n/a	✓
Parent-visible proof of thinking	✗	✗	Limited	Limited	✓

Key Insight: *The building blocks exist. What is missing is a product that makes AI teach thinking instead of replacing it, and can prove to a parent that it worked.*

6. Product Strategy

6.1 Primary User & Buyer

Role	Who	Core need	Product value
Buyer	Parent / guardian of a 5–17 child, already paying for preventive skilling	Evidence the child is thinking, not copying	Mastery evidence, honest safety reporting, short daily commitment
Primary user	The learner, in one of three bands	Something worth doing that ends in visible capability	Age-fit loop that always ends in feedback and a next step
Secondary user	Teacher (post-MVP)	See where a class is actually stuck	Misconception-level view, not marks

Product principle: the product only succeeds when the child does the thinking and the parent can see that they did.

6.2 Problem Statement

Children aged 5–17 are meeting general-purpose AI unsupervised and unstructured, in copy-the-answer mode, during the years their thinking habits form. Two failures recur: (1) the AI does the thinking before the child attempts anything, and (2) neither the child nor the

parent has any evidence of whether understanding occurred. Opportunity: replace the answer-first interaction with a retrieve-then-guide loop that produces provable mastery evidence, banded by age.

6.3 MVP Scope

Goal (unchanged from v1): take a learner from placement to a first provable mastery event — correct performance plus explanation or transfer — inside the first session.

Capability	MVP	Note
Guardian account, consent, and child account creation from /family	✓	Better Auth; roles ADMIN / STUDENT / TEACHER / GUARDIAN. A family holds up to four children and many guardians (P8).
Child login choice: internal skooling.in username, or Gmail for 13–17	✓	The internal address is a username with no inbox. Answers the community-versus-privacy split (P9).
Three-signal placement: prior knowledge, interest, ease/confidence	✓	Completes in under ten minutes and sets the band, the support level and the entry point on the graph.
Age-banded lesson flow: Subject → Topic → Lesson → ordered Blocks	✓	Block kinds: leaflet, question, story, repair, artifact. 4 subjects, 14 topics, 55 lessons authored.
Retrieve-then-guide loop: attempt → feedback → hint → retry → route update	✓	The core pedagogical contract, and the direct answer to the first interview finding.
Deterministic MCQ grading	✓	single_select and multi_select are graded in code and never delegated to a model.
Bounded text and picture judging against a rubric	✓	prompt and image_prompt questions are judged by the gated AI tutor against an authored rubric; schema-bound and auditable.
Misconception tagging and in-flow repair blocks	✓	A wrong option carries a misconception identifier and routes to the repair block written for that specific wrong idea.
Mastery as correct performance plus explanation or transfer	✓	Attempts, routing decisions and completion are persisted as learner state and drive what unlocks next.
Layered safety: screen every AI call, escalate, alert on severe	✓	One gated AI library, enforced by an ESLint rule plus a pre-commit hook. Full architecture in 7.5.
Parent visibility and the urgent alert path	✓	Per-child progress, try-at-home prompts, and tiered safety alert emails with the child's exact words.
Content packages authored by non-engineers, loaded by a validating importer	✓	node.json + assets/, validated against a shared schema at authoring time and again by a pre-commit hook.
Kid-safe AI image maker and family gallery	✓	5–7 build a picture from curated menus with no typing; 8–17 describe it in words. Sharing is guardian-gated twice over.

MVP flow: Consent → Band select → 3-signal placement → Tailored lesson → Attempt → Feedback/repair → Explanation → Mastery event → Parent summary

6.4 What We Will Not Build

Excluded	Rationale
Open free-chat AI tutor	It is the offloading mechanism and the safety exposure. Bounded roles only: text_judge, followup_helper, path_assist, safety_screen.
Homework help / answer generation	The behaviour parents told us they are trying to stop. Competing there means winning the wrong game.
Guided simulations in V1	Strong next candidate but needs a separate interaction engine. Deferred, not dropped.
Full open projects for 5–7	Readiness gate: projects mainly 13–17, guided builds 8–12, little or none at 5–7.
Streak-driven engagement mechanics	Rewards tie to mastery progress. Streaks must not become the product.
B2B / school sales, LMS integration	Consumer-parent wedge first. School procurement is a different product and a different cycle.
Subjects beyond AI literacy — revised	Revised since v1: maths, science and coding now run alongside AI on the same content model, importer and graph. AI remains the differentiator and the entry world; the others exist because a single subject could not carry a daily habit.

Scope principle: bound the AI and prove the learning. Everything else is later.

6.5 North Star Metric

Weekly Verified Mastery Events per Active Learner (WVME)

A Verified Mastery Event = a concept where the learner produced correct performance AND an explanation or transfer that passed the rubric. Completion alone never counts. Split into first-attempt mastery vs repair-then-mastery.

Why it holds both MVP problems	How
Thinking Evidence	The metric is only incrementable by evidence of understanding, so the product

	cannot win by shipping more content.
Bounded AI	A model that hands over answers inflates correctness but not explanation quality, so the split reveals it immediately.

Success signal: WVME rises while repair-then-mastery grows as a share — learners are being routed through their misconceptions rather than around them. Activation (from the decision log) = placement complete + first tailored lesson complete + first retry, follow-up or reflection event.

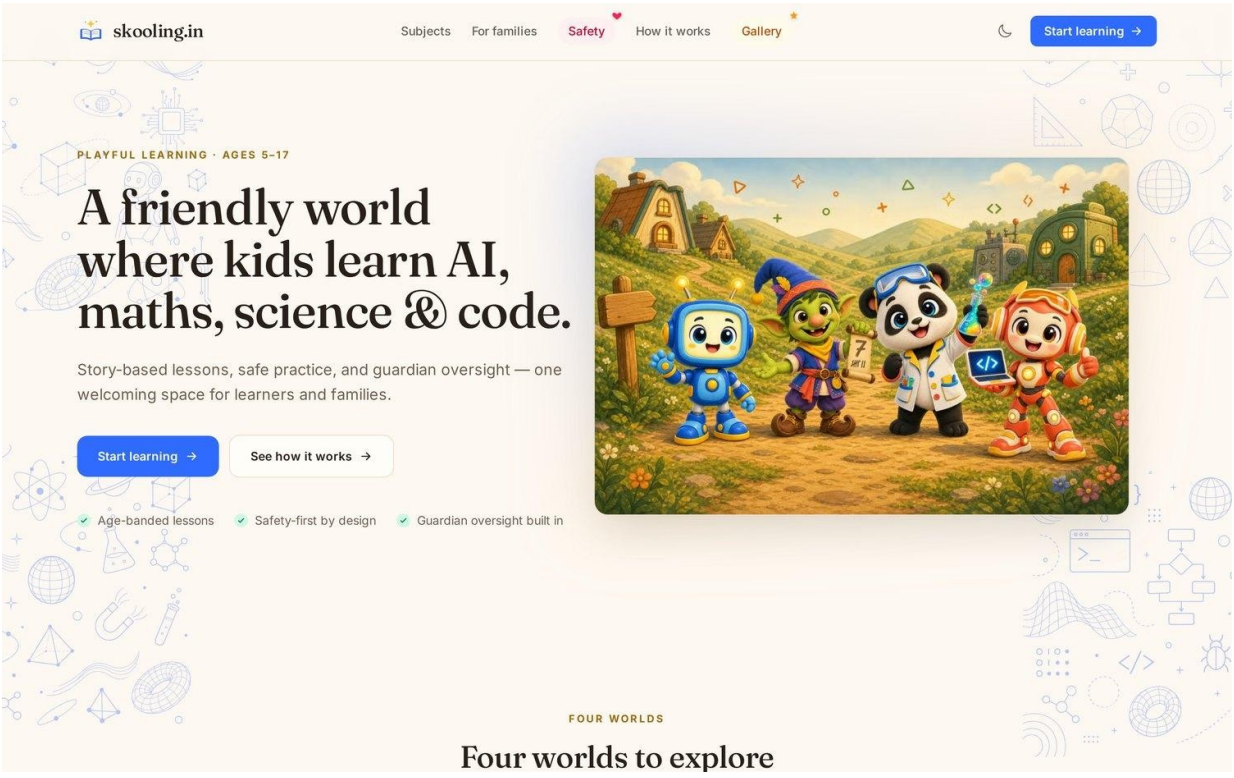
7. Solution Design

7.1 The redesigned loop

#	Step	What changes vs today
1	Placement — three signals in under 10 minutes	Routing uses readiness AND motivation, not a prior-knowledge test alone.
2	Retrieve — the learner attempts before anything is explained	Inverts the answer-first default. This is the anti-offloading mechanism.
3	Grade — deterministic for MCQ, rubric-bound for short text	Correctness is never a model's opinion where it can be computed.
4	Repair — a wrong answer routes to a misconception-specific repair block, in flow	Not "try again". The system knows which wrong model the child holds.
5	Explain — short self-explanation or transfer task (8–12, 13–17)	This is what converts a correct answer into a Verified Mastery Event.
6	Route — support level, review schedule and next node update from evidence	Scaffolds fade on evidence of success, never on a fixed lesson count.
7	Report — mastery evidence to the learner; summary to the parent	Parent gets proof of thinking instead of minutes-on-app.

7.2 Key screens by band

Band	Primary screen	Input	Support
5–7	Guided story-play: setup → choice/prediction → result → recap	Tap, drag, image choice; very low typing	Strong visual guidance, short steps, optional audio
8–12	Challenge-practice: retrieval check → guided build → reflect	MCQ, short bounded text, guided builds	Medium scaffolds, explicit retry, clear progress cues
13–17	Project-plus-reflection: predict → build → compare → critique	Text, comparison, projects, optional tools	Higher autonomy, strategic hints, deeper reflection
Parent	Daily summary: mastery events, misconceptions repaired, time on task	Read-only + controls	Urgent alerts only for harm signals



The public entry point at www.skooling.in. Four subject worlds, each with a mascot guide.

Cross-band state rules: preserve learner work through errors and network failure; show the next step after every answer; keep audio optional, never forced; respect reduced-motion preferences.

7.3 Integration with what is already built

Layer	Decision	Status
Auth & roles	Better Auth inside the Express API; ADMIN / STUDENT / TEACHER / GUARDIAN. A guardian creates the child's account from /family and picks the login mode.	Shipped
Runtime	Single Express 5 + TypeScript backend (apps/api) with internal modules, not microservices. React 19 + Vite frontend (apps/web). Both ship as one Docker image on Railway behind the www.skooling.in custom domain.	Shipped
Modules	onboarding, lesson delivery, grading, routing, follow-up, safeguarding, parent alert, analytics, content import, gallery, gamification	Shipped
Content graph	Neon Postgres; adjacency-list graph nodes and edges traversed with WITH RECURSIVE; lookup tables for node and edge types, so a new type is an insert rather than a migration.	Shipped
Content model	Subject → Topic → Lesson → Block[]. Lesson is the only linkable unit; prerequisite edges are Lesson ↔ Lesson and form a DAG.	Shipped
Authoring	Content packages (node.json + assets/) validated against a shared schema at authoring time and again by a pre-commit hook. New content of an existing type needs no engineering.	Shipped
Media	Lesson and UI artwork on a Cloudflare CDN in front of R2; children's generated pictures in a separate private bucket served only through short-lived signed links.	Shipped
Audit	Append-only events for attempts, routing, text judging, safety and parent alerts. Safety evidence is stored separately from learning analytics.	Shipped
Engineering gates	Seven pre-commit hooks: content validation, content coverage, CDN asset coverage, rich-web-content guard, typecheck, backend lint (which enforces the AI gate), and the test suite.	Shipped

7.4 Content model and who is allowed to grade

A lesson is an ordered list of blocks, not a page. The block kinds are the vocabulary the content team writes in, and they are the same across all three age bands — only the mix and the support level change.

Block kind	Role in a lesson
leaflet	Teach. Text, HTML or SVG with media.
story	A branching scene. Choices carry consequences rather than marks.
question	Ask. The graded check.
repair	Re-teach exactly one named misconception, then return the learner to the question.
artifact	A self-contained activity rendered in a sandboxed frame.

Correctness is never a model's opinion where it can be computed. The grading authority for each question type is a deliberate decision, not an implementation detail:

Question type	Graded by	Why
single_select	Deterministic code	The correct option is authored; a model adds risk and no accuracy.
multi_select	Deterministic code	Exact set match against the authored correct set.
prompt (written answer)	Gated AI judge, against an authored rubric	Judgement of an explanation cannot be computed. The judgement is schema-bound and every call is auditable.
image_prompt (make a picture with AI)	Gated AI judge, shown the lesson instruction and the finished picture	The judge is deliberately never shown the child's typed words — only the artwork and the task.

Every plausible wrong option carries a misconception identifier. A wrong answer routes down that edge to the repair block written for that specific wrong idea, then back to the question. The learner is not told "wrong, try again"; the system knows which wrong model they are holding.

Authored coverage — 4 subjects, 14 topics, 55 lessons, 100 question blocks. A question counts under every band its lesson targets, so band columns can exceed the distinct total.

Question type	5–7	8–12	13–17	Total
single_select	10	46	26	73
multi_select	0	5	4	6
prompt	0	2	13	15
image_prompt	2	4	2	6
All types	12	57	45	100

The delivery payload is sanitised: correctness flags, feedback, misconception identifiers, rubrics and sample answers are never sent to the learner's browser before grading.

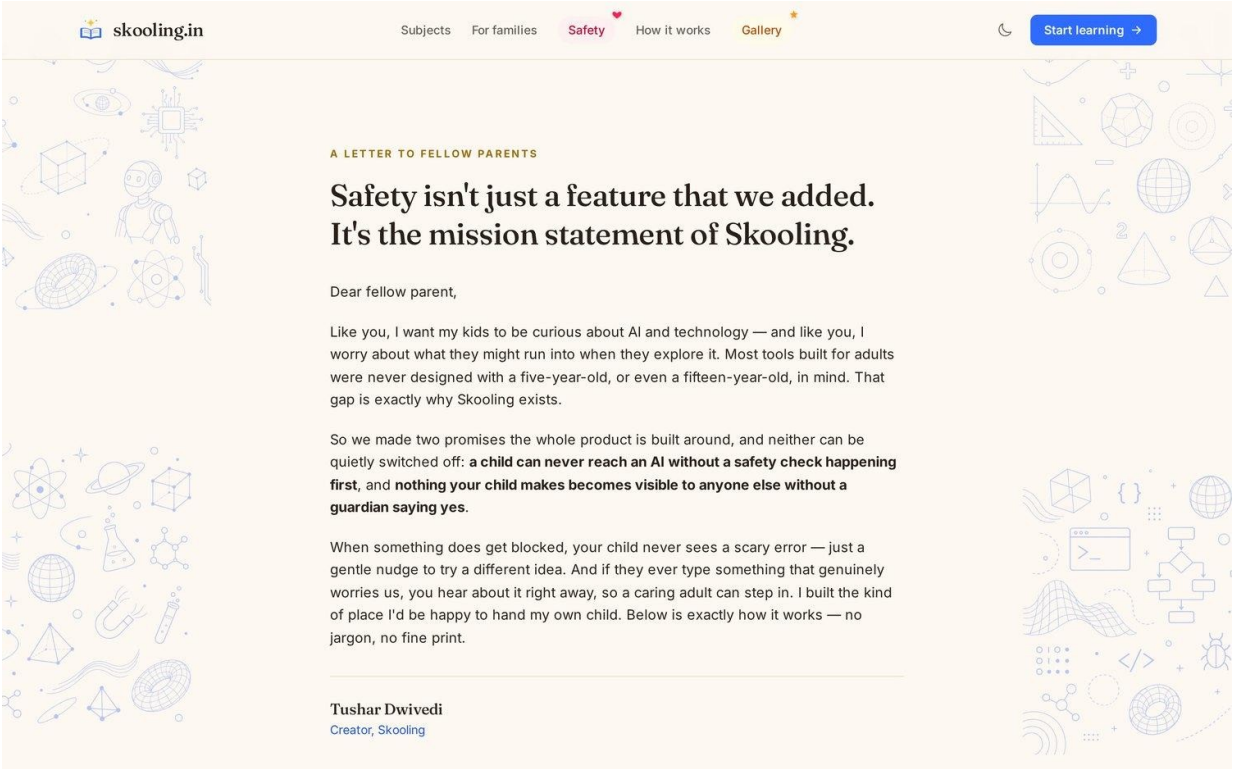
7.5 Safety and moderation architecture

Parents raised safety as an objection to the chat surface itself, not to any particular answer. The response is architectural rather than editorial: an AI cannot be reached without a safety check first, and the codebase makes any other path impossible to commit.

Decision	How it is enforced
One gated library is the only code allowed to call a model	An ESLint no-restricted-imports rule plus a pre-commit hook. A call that bypasses the gateway cannot be committed, so an unchecked call cannot exist in the deployed image.
Image generation is always check words → generate → check picture	The child's prompt text and the finished image are moderated independently, and the provider's own safety filter runs on top of both.
Ages 5–7 cannot type into AI at all	They compose a picture by choosing from curated menus, so a young child cannot author an unsafe request. With no age band set, the feature is unavailable — it fails closed.
The AI grader never sees the child's raw prompt	When the tutor judges a drawing it is shown the lesson instruction and the finished picture only, and its own written feedback is moderated before the child reads it.
Usage is capped	A daily cap on image generation per child, and a small number of graded tries per image question per day. Denied or errored attempts do not consume a try.
A block is kind, and never traps the child	Blocked requests show a warm message rather than an error, and neither a moderation block nor a technical failure stops the child finishing the lesson.

When a child's typed request is blocked for a serious reason, the response is tiered by severity rather than by volume convenience. The family email quotes the exact words the child typed, so a caregiver responds to the real thing rather than a summary. The operator is notified on every alert, and family and operator emails are always separate messages.

Flagged category	Alert	Family email	What the child sees
self-harm, self-harm/intent, self-harm/instructions	force	Urgent template, every single time. The rate limit is explicitly bypassed.	Compassionate wording — "I hear you. We all have bad days…"
sexual/minors	force	Every time, no throttle.	Firm — "That's not something we can make here. Let's try a different idea!"
sexual, hate/threatening, harassment/threatening, violence/graphic, illicit/violent	throttled	A gentler heads-up, de-duplicated to at most one per child per ten minutes.	Warm, generic wording.
harassment, hate, illicit, violence	none	No email.	The same warm, generic wording.



The public safety page — the same decisions, stated to parents in their own language.

7.6 Privacy, sharing and child-safe analytics

Interviews split on community: some families wanted their child to see and be seen by other learners, others wanted no exposure at all. The product does not pick a side. It gives the family two explicit choices and defaults both to the private option.

Choice	The options	Default
How a child signs in	An internal skooling.in username, which is a login with no inbox and cannot receive mail; or a real Gmail account, available for ages 13–17 only.	Internal username
How a shared picture appears	Named, or anonymous, chosen by the guardian at the moment they approve the individual picture.	Anonymous

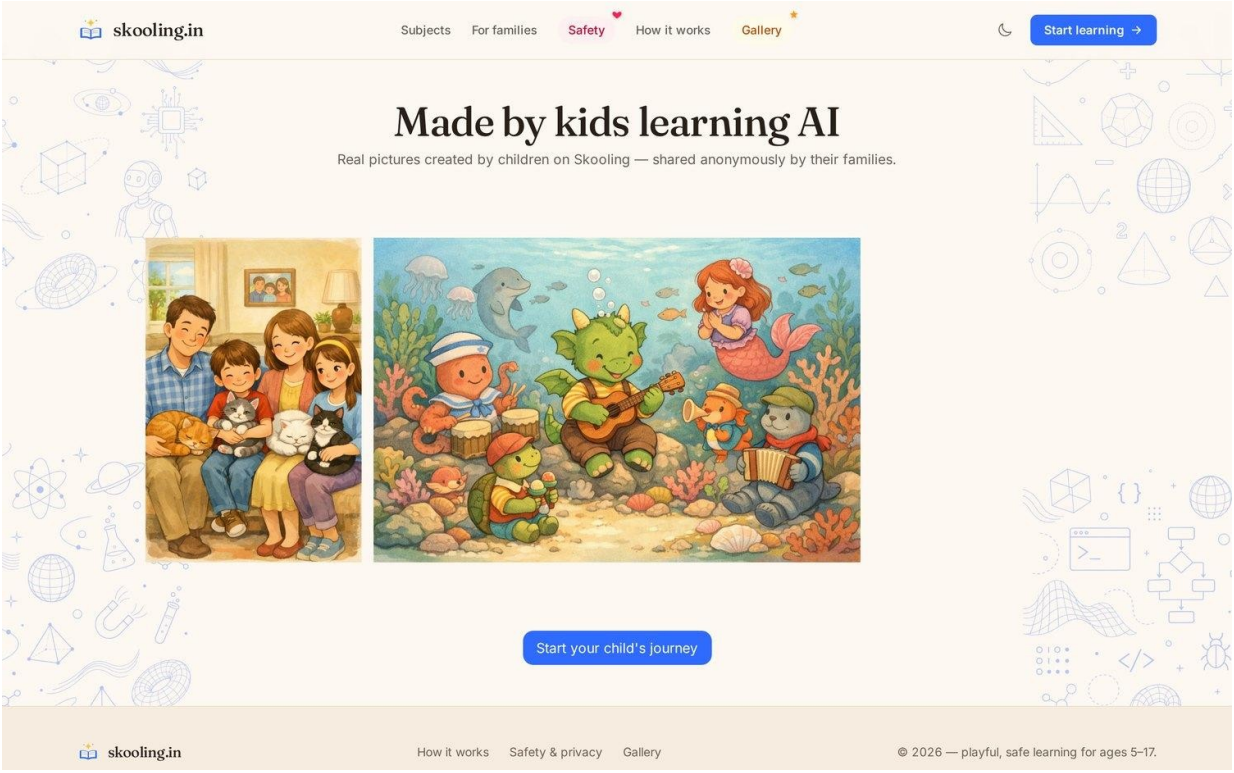
Nothing a child creates becomes visible to anyone else without a guardian acting twice. The ladder is deliberate:

Step	Who acts	Rule
1 · Unlock sharing	Head guardian	A one-time family consent. Until it is given, nothing can be shared at all.
2 · Request	Child	A child can only ask. They can never publish. Duplicate requests for the same picture collapse, so a parent's queue cannot be flooded.
3 · Approve and scope	Head guardian	Per individual picture, with the audience chosen: family, local community, or public — and named or anonymous.
4 · Publish	System	The public copy is a separate file with its image metadata stripped, so nothing embedded travels with it.
5 · Reverse	Any guardian, at any time	Un-sharing removes the public copy immediately; deleting removes the picture from storage entirely. Anyone, signed in or not, can report a shared image into an admin takedown queue.

Measurement is subject to the same principle. A children's product should not be able to answer questions about a specific child inside a third-party analytics tool, so it cannot:

Rule	What it means in practice
A child is identified only by a one-way hash	The distinct identifier sent to analytics is an HMAC-SHA256 of the child's database id, keyed by a server-only secret. It is deterministic, so a child's events still group; it cannot be reversed, and the real id never leaves the server.
Events only — no people profiles for children	No persistent, human-identifiable profile for a child may exist in the analytics system. Adults are deliberately different: guardians and teachers use ordinary full identity.
Two properties, and no more	Age band and subject are the only permitted child properties. No names, no email addresses, no free text, no answers, no image prompts.
Safety evidence never enters analytics	Flagged text, moderation categories and self-harm specifics live only in the application's own safety store. Analytics is told that a screening happened and at what level — never who, and never what.
One anonymous macro aggregate	A single screening event carrying age band, outcome, alert level and surface. Enough to run a safety programme; not enough to identify a child.

Trade-off accepted: *We gave up per-child analytical depth — retargetable profiles, rich segmentation, the ability to reconstruct one child's journey — in exchange for a system that structurally cannot profile a minor.*



The public gallery. Every picture was approved by a guardian, and appears named or anonymous by their choice.

7.7 Design decisions

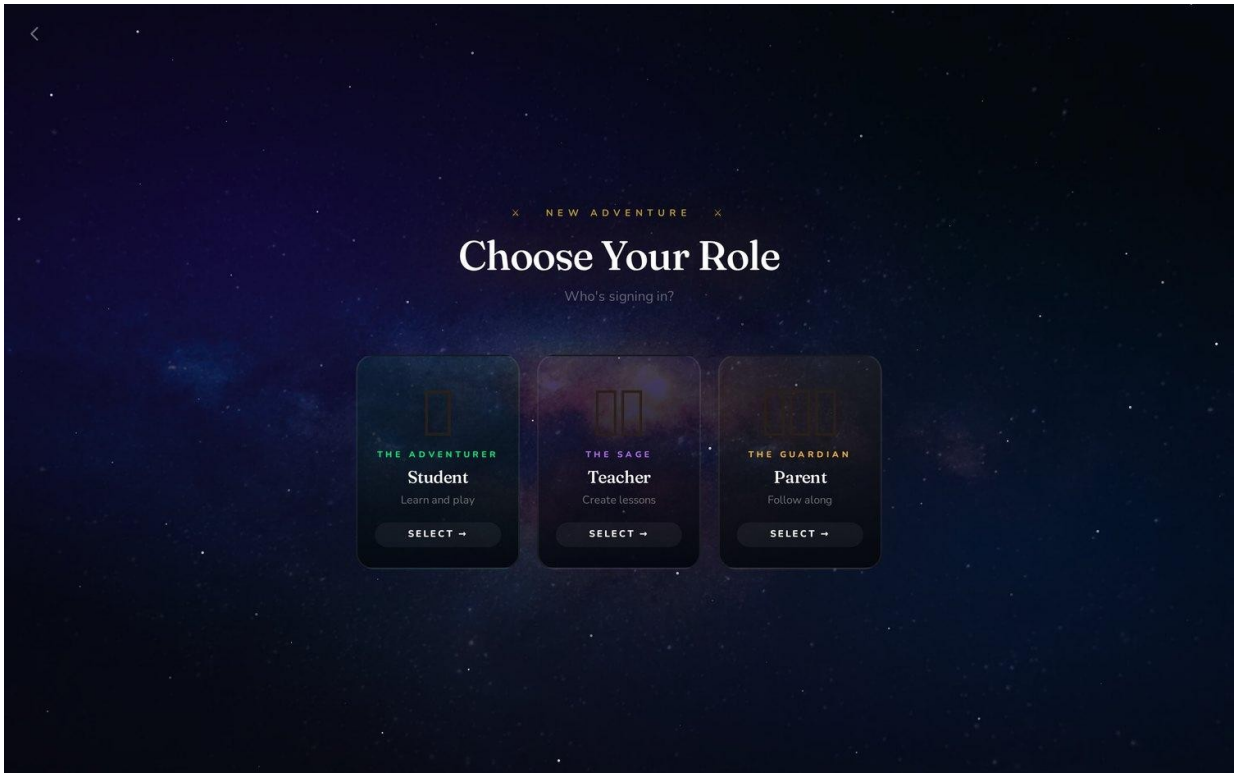
One codebase carries two visual systems, and the split is deliberate. A six-year-old exploring a world and a parent checking progress at eleven at night are not the same design problem, and one system stretched across both would serve neither.

	The child world	The adult surfaces
Treatment	Bespoke and art-directed — space, ocean maps, quest frames, a mascot who speaks each card.	Calm, dense, credible utility interface for marketing, parent, teacher and admin.
Theming	Deliberately not theme-toggled. It is art direction, like a game; a toggle would fight the art and add confusion rather than value.	Light and dark from a single stylesheet of semantic tokens, so re-theming is a handful of edits rather than thousands of class names.

Why	Ages 5–7 cannot read fluently, so the interface is tap, drag and picture, with optional audio and no typing into AI.	Readability, density and night use genuinely matter to the buyer.

Light is the default and dark is an explicit opt-in, stored per browser and deliberately not inferred from the operating system. Learning is text-heavy reading, and extreme white-on-black harms dyslexic and astigmatic readers; dark exists for the adult working late, as their choice. We do not claim dark mode is better for the eyes — that is a myth, and the honest claim is comfort and preference.

Accessibility commitments: WCAG AA contrast on body text; colour is never the only signal; and every ambient animation is gated behind the operating system's reduced-motion preference. That last one came from a parent describing how fast children's television moves — a comfort and attention accommodation for motion-sensitive children, not a developer nicety.



The immersive child-facing entry. The adult surfaces of the same product are calm and themeable; this one is art-directed and never toggles.

8. Metrics & Success

Tier	Metric	Definition
North Star	Weekly Verified Mastery Events per active learner	Correct performance + passing explanation/transfer, per learner per week
Product	Activation rate	Placement complete + first tailored lesson complete + first retry/follow-up/reflection event, in session one
Product	Mastery rate by topic	Share of learners reaching mastery, by band — not content completed
Product	Repair effectiveness	Share of tagged misconceptions that reach mastery after a repair block
Retention	D1 / D7 / D30 by cohort	Segmented by age band, onboarding path, and reward-system exposure
Business	Paid conversion and monthly retention	Against a parent-buyer benchmark, not an app-engagement benchmark
Trust	Parent trust signal	Summary open rate, controls usage, complaint rate
Safety	Safety precision	Share of high-risk alerts still valid after review

Instrumentation: lesson starts and completions, placement completion, image creation, share requests and approvals, child and guardian activation, progress views, and one anonymous safety-screening aggregate — all emitted server-side through a single seam, with children pseudonymised as described in 7.6.

Guardrail metrics

Guardrail	Watching for
Hint-before-attempt rate	The product re-creating offloading inside its own loop.

Correct-without-explanation gap	Rising correctness with flat explanation quality = answers are coming from somewhere else.
Activity-without-mastery	Rewards driving usage that does not convert to mastery — the streak failure mode.
Alert volume vs precision	Alert fatigue destroying parent trust faster than misses would.
Session length by band	5–7 sessions drifting past attention span.

Success definition: a child completes a session having attempted before being told, been routed through at least one misconception, explained one idea in their own words — and the parent can see it. If WVME rises while hint-before-attempt falls, the product is working.

9. Market Sizing (TAM / SAM / SOM)

Sized bottom-up on users, India-first. Every input is tagged; the assumptions are stated so they can be attacked. Figures previously circulating in our team draft were re-verified and three did not survive — see the correction note below.

Level	Definition	Size	Basis
TAM	All Indian children in Classes 1–12 whose parents could buy a supplemental learning product	≈ 233M students (24.69 crore total enrolment incl. pre-primary; 1.40 crore pre-primary)	UDISE+ 2024-25, Ministry of Education [V]. 1,471,473 schools. Note: enrolment is declining, not growing — do not model a growing base.
SAM	Children 5–17 in households with home internet and an existing supplemental-learning spend	Funnel, not a single number: 233M × 86.3% household internet × (private-school / paying-household share)	Household internet on premises 86.3% (rural 83.3%, urban 91.6%) — MoSPI Comprehensive Modular Survey: Telecom, May 2025 [V]. School internet is only 63.5% (UDISE+) [V]. Paying-household share is NOT YET VERIFIED — this is the weakest link and must be sourced from NSS/ASER before submission.
SOM	3-year obtainable: English-medium tier-1/2 households already buying coding or abacus, reached via parent communities and schools	Model at 150–300K children	ARPU must be re-based: posted prices are ₹600–1,200/month for group abacus and ₹2,000–2,500/month for branded abacus or 1-on-1 coding [V, vendor price lists]. At ₹800/month ARPU, 200K children ≈ ₹192 Cr/yr [A].

Corrections to figures circulating in the team draft

Claim as drafted	Status	What to use instead
"~250M K-12 students"	Overstated	≈233M in Classes 1–12; ~247M including pre-primary (UDISE+ 2024-25) [V]
"85–90M students have internet access"	Contradicted	No student-level source exists. Use household internet 86.3% (MoSPI 2025) or school internet 63.5% (UDISE+), and say which [V]
"Abacus market €2.7B → €4.3B; 3M+ enrolments/yr"	Do not use	Single consultancy blog, no methodology; the €2.7B/€4.3B figure is a tier-I/II segment, and its companion total (€14.2B) exceeds India's entire projected 2030 edtech market. The 3M enrolment figure has no traceable source at all.
"NEP 2020 introduced coding from grade 6"	Overstated	NEP 2020 ¶4.25 recommends "activities involving coding" at the Middle Stage (Grades 6–8). Not a mandated subject [V]
"₹500/month ARPU"	Too low by 2–5×	₹500 is roughly what one class costs. Realistic monthly ARPU: ₹600–1,200 budget, ₹2,000–2,500 branded [V]
"China mandated AI education nationwide from Sept 2025"	Partially wrong	Beijing is the first province-level region to make it compulsory from Sept 2025 (1,400+ schools, 8+ class hours/grade); the national goal is to universalise by 2030 [V]

Key Insight: *The strategic point survives the corrections: the pain is latent for most 5–12 parents, so we are not creating a budget line — we are redirecting a proven preventive-skilling spend. But the revenue model must be re-run at a realistic ARPU and the paying-household share must be sourced before this section is defensible.*

10. Go-To-Market

Phase	Motion	Why
0–3 mo · Wedge	8–12 band only, English-medium, tier-1/2 India. Parent communities, school WhatsApp groups, one flagship AI-literacy topic.	8–12 is the intervention window: habits not yet hardened, and the child can already reflect on their own thinking.
3–9 mo · Proof	Publish the mastery-evidence report as the marketing asset. Parent referral loop driven by the weekly summary, not by streaks.	The summary IS the acquisition mechanism — it is the only artefact that shows a parent what they bought.
9–18 mo · Extend	Add 13–17 (projects, critique) then 5–7 (story-play, audio-first). Pilot a teacher view once teacher interviews land.	Each band reuses the same graph and importer; only the block mix and support level change.
Pricing	₹800–1,200/month, positioned against one abacus or coding class, not against a free chatbot	[A] Must be validated. Anchor on the class the parent gives up, not on software comparables.

11. Risks & Trade-offs

Risk	Why it matters	Mitigation / trade-off accepted
Latent pain in 8–12 parents	Most parents of 8–12s do not yet feel the dependency their 13–17 counterparts describe. We are selling prevention.	Lead with the visible artefact (mastery evidence), not with the fear. Accept a longer education-led sales cycle.
Refusing homework help caps adoption	It is what children actually want and what free tools already do well.	Deliberate. Competing on homework means winning the behaviour we exist to prevent. Trade reach for defensibility.
Rubric judging is a model, and models drift	Mastery evidence loses meaning if the judge is unreliable.	Schema-bound prompts, deterministic grading wherever computable, auditable text_judge events, and the correct-without-explanation guardrail.
Safeguarding failure is existential on a minors product	One mishandled severe case ends the company.	Layered screening on every text event, human review on severe, separate safety storage, data minimisation (UNICEF, Common Sense).
Three bands triple content cost	5–7 and 13–17 need genuinely different objects, not restyled ones.	Ship 8–12 first. The block model and importer are band-agnostic; only the block mix and support level differ.
Market data is analyst-grade	Coding and AI-in-K-12 sizes come from vendor press releases with a 14–18.6% CAGR spread.	Attributed as estimates, never asserted. TAM is built on government census data instead.
Evidence gaps in our own research	Teacher interviews pending; professional quotes in the team draft are simulated placeholders; n=6 on parents/children.	Marked explicitly throughout. Teacher interviews and a larger parent sample before any funding conversation.

12. References

Primary research

6 completed interviews (5th-grade children, post-10th-grade children, and their parents), Google Forms survey, and professional interviews across teaching, sales, operations and IT. Quotes marked [P] are verbatim. Teacher interviews pending. Professional quotes in the team draft are simulated placeholders and are excluded from this document.

Verified secondary sources [V]

- Gerlich, M. (2025). AI Tools in Society: Impacts on Cognitive Offloading and the Future of Critical Thinking. Societies 15(1), 6. Correction: Societies 15(9), 252.
- Sparrow, Liu & Wegner (2011). Google Effects on Memory. Science 333, 776.
- Common Sense Media (Sept 2024). The Dawn of the AI Era: Teens, Parents, and the Adoption of Generative AI at Home and School.
- OECD (2024). PISA 2022 Results (Volume III): Creative Minds, Creative Schools.
- Executive Order 14277, Advancing Artificial Intelligence Education for American Youth (23 April 2025).
- Ministry of Education, PRC — Notice on Strengthening AI Education in Primary and Secondary Schools; China Daily (Oct 2025) on Beijing's compulsory rollout.
- Ministry of Education, India — UDISE+ 2024-25; National Education Policy 2020 (¶4.23, ¶4.25, ¶4.4).
- MoSPI — Comprehensive Modular Survey: Telecom 2025 (household internet access).
- IES — Organizing Instruction and Study to Improve Student Learning; EM2; EPIGAME; Comprehension SEEDING.
- APA — Practice for Knowledge Acquisition; Misconceptions; Motivation.
- UNICEF — Policy Guidance on AI for Children; Formative Assessment for Quality, Inclusive Digital and Distance Learning; Tinkering with Tech and AI.
- Brookings — Realizing the promise: How can education technology improve learning for all?
- W3C WAI — Input: typing, writing, and clicking; WCAG 2.2 G79 (spoken version of text).
- Smashing Magazine — Designing Web Interfaces For Kids (2015); A Practical Guide To Designing For Children (2024).
- Common Sense — Kids Tech Privacy. Mind the Product — retention and streak design. Laws of UX — Goal-Gradient Effect.

Attributed estimates [S]

Spherical Insights (online coding-for-kids market); market.us and Polaris (AI in K-12 market); vendor price lists (Bhanzu, SIP Abacus, ALLEN IntelliBrain) for observed tuition pricing. These are analyst or commercial figures and are attributed, not asserted.

Internal working references

docs/grill-me/01–15 (session brief, age bands, business & product, UX, learning science, content & interactives, learning graph, AI systems, safeguarding, backend architecture, gamification, decisions, open questions, references, implementation checklist) · docs/content-model/README.md · docs/learningplatform-team-research-report.docx · Cohort group PRD (teammate draft, content reference).

Internal sources of truth for v2 (code and repository documents, not summaries): docs/features.md · docs/child-safety-and-privacy.md · docs/safety-alert-policy.md · docs/analytics-plan.md · docs/mixpanel-project.md · docs/design-decisions.md · docs/content-coverage.md (generated) · content/QUESTION-SCHEMA.md · docs/grill-me/01–15 · CLAUDE.md.